

ABM BUSINESS LOGIC BIBLE

DEEP COHERENCE AUDIT

Cross-Section Conflict Analysis Across All 19 Sections

V1.3 FSD + V2 Deep Specification | Decimal Technologies

AUDIT SCOPE AND METHODOLOGY

This audit systematically cross-references every rule, threshold, timing window, state transition, decision gate, and architectural constraint across all 19 sections of the ABM Business Logic Bible (V1.3 FSD base document + V2 deep-specification layers built across Sections 1-16).

Every conflict identified traces to a specific engine failure mode: deadlock, contradictory automation, data corruption, capital waste, or governance collapse. Conflicts are classified by severity:

- CRITICAL** - Engine will produce incorrect behavior, deadlock, or data corruption. Must resolve before implementation.
HIGH - Engine will produce suboptimal behavior, wasted capital, or governance gaps. Must resolve before production.
MEDIUM - Engine will function but with ambiguity that creates implementation risk. Resolve during engineering review.

AUDIT FINDINGS SUMMARY

Total Conflicts Identified: 34

- CRITICAL (engine-breaking): 11
HIGH (governance/capital risk): 14
MEDIUM (ambiguity/implementation risk): 9

- Conflict Categories:
- A. Anchor Model Conflicts (Account vs Opportunity) - 5 conflicts
 - B. Scoring and Tier Conflicts - 6 conflicts
 - C. State Machine and Lifecycle Conflicts - 5 conflicts
 - D. AI Agent and Autonomy Conflicts - 4 conflicts
 - E. Timing and Sequence Conflicts - 4 conflicts
 - F. Integration and Data Flow Conflicts - 4 conflicts
 - G. Governance and Learning Loop Conflicts - 3 conflicts
 - H. KSA-Specific and Compliance Conflicts - 3 conflicts

CATEGORY A: ANCHOR MODEL CONFLICTS

The single most architecturally dangerous category. V2 Section 1 re-anchored the unit of selling from Account to Opportunity (Entity 38). But V1.3's entire pipeline, database schema (Section 19), state machine (Section 23), scoring engine (Section 4), and CRM sync (Section 9/12) all assume Account is the primary entity. Every downstream section inherits this contradiction unless explicitly resolved.

[CRITICAL] COH-A01: Account-Centric vs Opportunity-Centric Design

Sections: V2 Section 1 (Meta-Principles) vs V1.3 Principle 2 vs V1.3 Section 19 (DB Schema) vs V1.3 Section 23 (State Machine)

Conflict Description: V2 Section 1 establishes Opportunity (E38) as the unit of selling with Account as the relationship container. V1.3 Principle 2 explicitly states 'The platform is not lead-centric. It is account-centric. All workflows revolve around ACCOUNT.' V1.3 Section 19 Entity Relationship Model shows ACCOUNT as the root with all other entities as children. V1.3 Section 23 defines account states (WATCHING/ENRICHING/HOT/OUTREACH_ACTIVE/etc.) as the primary state machine.

Rule A: V2 Section 1: 'Opportunity (Entity 38) is the unit of selling. Account is the relationship container.'

Rule B: V1.3 Principle 2: 'All workflows revolve around ACCOUNT.' + V1.3 Section 19: ACCOUNT as root entity.

Engine Impact: The scoring engine scores Accounts, but the decision engine should decide on Opportunities. Enrichment triggers on HOT Accounts, but value is created on Opportunities. Multiple Opportunities per Account are architecturally possible but the entire pipeline assumes one-to-one. The state machine tracks Account states, not Opportunity states.

Recommended Resolution: RESOLVE BY: Implement Hybrid Path Option C+ (already decided in OPEN-Q14.3). Account remains the scoring and enrichment anchor for MVP. Opportunity entity is created when the engine decides to pursue, inheriting the Account's score and enrichment. The state machine must be DUAL: Account states (relationship-level) AND Opportunity states (deal-level). The DB schema needs an Opportunities table with account_id as FK. All downstream sections must specify whether they operate on Account or Opportunity.

[CRITICAL] COH-A02: Scoring Anchor: Score Belongs to Account or Opportunity?

Sections: V2 Section 4 (Scoring Engine) vs V2 Section 1 (Opportunity Re-anchoring) vs V1.3 Section 4

Conflict Description: V2 Section 4 specifies $TOTAL_SCORE = Signal_Strength \times 0.35 + Regulatory_Pressure \times 0.30 + Persona_Reachability \times 0.20 + Relationship_Warmth \times 0.15$ — computed at the ACCOUNT level. But if Opportunity is the unit of selling, which entity does the score belong to? An Account can have multiple Opportunities at different stages. The score's four dimensions are all Account-level properties, not Opportunity-level.

Rule A: V2 Section 4: Score computed per Account with four Account-level dimensions.

Rule B: V2 Section 1: Opportunity is the unit of selling. Decision Score = $f(Effective_Opportunity)$.

Engine Impact: If the score is Account-level but decisions are Opportunity-level, the engine cannot correctly route two Opportunities from the same Account at different stages (e.g., one in enrichment, one in outreach). The Effective Opportunity equation from Section 1 multiplies by Stage Bonus and Window Multiplier — these are Opportunity-specific, not Account-specific. The two scoring models are incompatible at the entity level.

Recommended Resolution: RESOLVE BY: The composite score is Account-level (it measures account attractiveness). Effective Opportunity is computed per-Opportunity, inheriting the Account's composite score plus Opportunity-specific modifiers (Stage Bonus, Window Multiplier, Budget Modifier, Entrenchment Modifier). This means the Section 4 score is an INPUT to the Section 1 equation, not a replacement. Both must coexist. The Section 4 score triggers tier assignment; the Effective Opportunity score governs resource allocation within a tier.

[CRITICAL] COH-A03: Tier Assignment: Account Tier vs Opportunity Tier

Sections: V2 Section 5 (Tier Assignment) vs V2 Section 1 (Opportunity Anchor)

Conflict Description: V2 Section 5 specifies HOT/WARM/COLD tiers assigned to ACCOUNTS based on the composite score. But if an Account has two Opportunities — one HOT, one COLD — what is the Account's tier? The tier gates enrichment and outreach (HOT triggers enrichment, COLD gets monitor-only). If the Account is rated by its best Opportunity, COLD Opportunities on HOT Accounts get free enrichment. If rated by its worst, HOT Opportunities on COLD Accounts get blocked.

Rule A: V2 Section 5: Tier assigned per Account. HOT $\geq$ 75 triggers enrichment + outreach.

Rule B: V2 Section 1: Opportunity is the decision unit. Multiple Opportunities per Account possible.

Engine Impact: Engine deadlock: An Account with one active Opportunity in OUTREACH_ACTIVE and a new signal suggesting a second Opportunity cannot be correctly routed. The tier system has no concept of per-Opportunity routing.

Recommended Resolution: RESOLVE BY: Account tier = MAX(Opportunity tiers) for relationship-level decisions (e.g., how we treat the Account overall). Opportunity tier = its own Effective Opportunity score for execution decisions (enrichment, outreach). Add TIER-MULTI-001 rule: 'When an Account has multiple Opportunities, the Account tier reflects the highest-tier Opportunity. Workflow routing operates on Opportunity tier, not Account tier.'

[HIGH] COH-A04: CRM Sync: HubSpot Deal vs Account Object Mapping

Sections: V1.3 Section 9 (Human Handoff) vs V1.3 Section 12 (CRM Sync) vs V2 Section 12 (Distributed Reality Coordination)

Conflict Description: V1.3 Section 9 specifies: 'Create CRM Opportunity' as a handoff action. V1.3 Section 12 lists 'POST /sync-hubspot' to synchronize 'opportunities and engagement.' But V1.3's entire pipeline up to this point has no Opportunity object — it only has Accounts. When does the CRM Opportunity get created? V2 Section 12 treats CRM as a 'reality coordination' system but doesn't specify when the Opportunity entity (E38) materializes in HubSpot vs the internal database.

Rule A: V1.3 Section 9: Create CRM Opportunity on positive reply.

Rule B: V2 Section 1: Opportunity is the unit of selling from inception (not just at handoff).

Engine Impact: CRM becomes out of sync with the internal database. Internal Opportunity exists before CRM Opportunity. Sales team sees Account in CRM but no Opportunity until reply — losing the strategic context the engine has built.

Recommended Resolution: RESOLVE BY: Define two Opportunity lifecycle stages: (1) Internal Opportunity created when engine decides to pursue (at HOT tier assignment or enrichment trigger), (2) CRM Opportunity created at MEETING_BOOKED or positive reply (whichever comes first). The internal→CRM sync rule must specify which fields transfer and when.

[HIGH] COH-A05: Engagement Tracking: Events Belong to Contact, Account, or Opportunity?

Sections: V1.3 Section 8 (Engagement Tracking) vs V1.3 Section 19 (DB: engagement_events table) vs V2 Section 1

Conflict Description: V1.3 Section 8 tracks engagement events (opens, clicks, replies) but doesn't specify the parent entity. V1.3 Section 19's engagement_events table maps to account/contact. V2 Section 1 says Opportunity is the unit — so engagement should ladder up to Opportunity, not just Account. But a Contact can be part of multiple Opportunities, and an engagement event (e.g., email open) doesn't carry Opportunity context.

Rule A: V1.3 Section 8/19: Engagement maps to Account + Contact.

Rule B: V2 Section 1: Opportunity is the unit of selling. Engagement should measure Opportunity progress.

Engine Impact: Engagement scores cannot be correctly attributed to Opportunities. A reply from a Contact on Opportunity A would also register on Opportunity B if the Contact is in both buying committees.

Recommended Resolution: RESOLVE BY: Engagement events carry triple-key: contact_id + account_id + opportunity_id. The opportunity_id is inferred from the outreach sequence that generated the message (each sequence is tagged with its Opportunity). For inbound engagement (Contact replies unprompted), the engine routes to the most-recently-active Opportunity.

CATEGORY B: SCORING AND TIER CONFLICTS

The scoring engine is the brain that gates every downstream action. Conflicts here cascade into enrichment, outreach, and capital allocation failures.

[CRITICAL] COH-B01: Two Competing Scoring Models: V1.3 Composite vs V2 Effective Opportunity

Sections: V1.3 Section 4 vs V2 Section 1 (Master Equations) vs V2 Section 4

Conflict Description: V1.3 Section 4 defines a single composite score: $TOTAL = Signal(35\%) + Regulatory(30\%) + Reachability(20\%) + Relationship(15\%)$. V2 Section 1 defines a completely different equation: $Effective\ Opportunity = Dynamic_Score \times ICS/100 \times Stage_Bonus \times Budget_Modifier \times Entrenchment_Modifier \times Risk_Modifier \times Window_Multiplier$, with $Decision_Score = Effective_Opportunity \times Strategic_Readiness/100 / Capital_Cost$. V2 Section 4 attempted to bridge these by saying 'V2 composite is MVP, Effective Opportunity is architectural successor.' But the transition criteria (EPIS-RCM-04 production maturity bar) are not specified concretely.

Rule A: V1.3 / V2 Section 4: $TOTAL_SCORE$ = weighted sum of 4 dimensions (0-100 scalar).

Rule B: V2 Section 1: $Decision_Score$ = multiplicative chain of 8+ factors (unbounded output range).

Engine Impact: Two different score types with different scales, different semantics, and different entity anchors. An engineer implementing Section 5 (Tier Assignment) doesn't know which score to threshold against. The 75/50 thresholds are calibrated for the 0-100 composite score, not for the multiplicative $Decision_Score$.

Recommended Resolution: RESOLVE BY: Explicit dual-score architecture. $Score_Composite$ (Section 4) produces a 0-100 scalar per Account, used for tier assignment (Section 5). $Score_Effective_Opportunity$ (Section 1) produces a value per Opportunity, used for resource allocation and prioritization within a tier. Both scores are computed and stored. Section 5 thresholds apply ONLY to $Score_Composite$. The EPIS-RCM-04 transition point must specify: when all reasoning streams (SIG-VSAT, SIG-RELEVANCE, ACCESS_STRENGTH) have confidence ≥ 0.7 for ≥ 30 days, $Score_Effective_Opportunity$ becomes the primary sort key within each tier.

[HIGH] COH-B02: Hysteresis Band Values: Section 5 vs Section 4 Recalibration

Sections: V2 Section 5 (Tier Assignment) vs V2 Section 4 (Score Recalibration) vs V1.3 Section 15.3

Conflict Description: V2 Section 5 specifies asymmetric hysteresis: enter HOT ≥ 78 , leave HOT ≤ 70 (8-point band). Enter WARM ≥ 53 , leave WARM ≤ 47 . But V2 Section 4 (SCORE-RECAL-001) says weights can be adjusted by $\pm 5\%$ per dimension per recalibration cycle. A 5% weight shift on Signal Strength (35% $\rightarrow$ 30%) changes the composite score by up to 5 points — potentially larger than the hysteresis band, making the band ineffective after recalibration.

Rule A: V2 Section 5: Hysteresis band = 8 points (70-78) for HOT.

Rule B: V2 Section 4: Weight recalibration can shift scores by up to 5 points per dimension.

Engine Impact: After a recalibration cycle, the hysteresis band may no longer prevent flicker. Accounts that were stable-HOT could suddenly drop to WARM not because anything changed about the account, but because the weights shifted.

Recommended Resolution: RESOLVE BY: Add SCORE-RECAL-007: 'After any weight recalibration, all accounts within 10 points of any tier boundary must be re-evaluated with a 48-hour tier-freeze. No tier changes take effect until the freeze expires and the recalibrated score is confirmed stable.' Also add: weight recalibration must not shift any single account's score by more than the hysteresis band width in a single cycle.

[HIGH] COH-B03: Signal Freshness Window: 7-Day Cutoff vs Decay Multiplier Contradiction

Sections: V1.3 Section 3.1.4 (Duplicate Detection) vs V2 Section 4 (Signal Freshness Multipliers) vs V2 Section 5 (Fresh-Signal Gate)

Conflict Description: V1.3 Section 3.1.4 says 'If signal already exists within freshness window, ignore signal.' This implies a binary fresh/stale model. V2 Section 4 specifies graduated freshness multipliers (FRESH 0-7 days = 1.0, RECENT 8-14 = 0.7, AGING 15-21 = 0.4, STALE 22+ = 0.1 per EPIS-HALF). V2 Section 5 says 'HOT is reserved for accounts whose score includes at least one FRESH (0-7 days) signal.' Three different freshness models operating simultaneously.

Rule A: V1.3: Binary fresh/stale with 'freshness window' (undefined duration).

Rule B: V2 Section 4: 4-tier graduated decay. V2 Section 5: Binary FRESH gate for HOT tier.

Engine Impact: The duplicate detection freshness window is undefined. If set to 7 days, it matches V2's FRESH definition but would block legitimate updates to the same signal type. If set longer, duplicates suppress scoring updates. The three models produce different outcomes for the same signal.

Recommended Resolution: RESOLVE BY: Separate the concepts. Duplicate detection window = signal_type + source + account + 48 hours (prevents ingestion spam). Freshness decay = V2's 4-tier model (governs score contribution). Fresh-signal gate = at least one signal with age <= 7 days (governs HOT eligibility). These are three independent mechanisms operating on different aspects of signal management.

[HIGH] COH-B04: SCORE-005 Engagement Update vs SCORE-RECAL-001 Monthly Recalibration: Frequency Conflict

Sections: V1.3 Section 4.5 (SCORE-005) vs V2 Section 4 (SCORE-CALC-003) vs V2 Section 4 (SCORE-RECAL-001)

Conflict Description: SCORE-005 says 'System shall retrigger scoring on engagement updates.' SCORE-CALC-003 says score recalculates on 'new signal, engagement event, enrichment update, relationship change, or scheduled monthly recalibration.' But SCORE-RECAL-001 says 'After 90 days of operation with at least 10 meetings booked, analyze and adjust weights.' Question: does engagement-triggered rescoring use the CURRENT weights or wait for the 90-day recalibration? If engagement events trigger rescoring 50 times/day across all accounts, is that sustainable?

Rule A: SCORE-005 + SCORE-CALC-003: Event-driven scoring on every engagement event.

Rule B: SCORE-RECAL-001: Weight changes only after 90 days with 10+ meetings.

Engine Impact: High-frequency rescoring with static weights produces identical results after the first engagement event in a cycle — wasting compute. Weight recalibration is too infrequent to capture early-stage learning. No rate limiting on scoring events means a viral LinkedIn post generating 200 profile views triggers 200 rescoring events.

Recommended Resolution: RESOLVE BY: Add scoring debounce: SCORE-CALC-004: 'Engagement events are batched per account with a 4-hour debounce window. Multiple events within the window are aggregated into a single rescore.' For recalibration: reduce to 30-day cycles after MVP with 5+ meetings as the minimum data threshold.

[MEDIUM] COH-B05: Signal Priority P1/P2/P3 vs Scoring Points: Inconsistent Value Assignment

Sections: V1.3 Section 3.1.2 (Signal Types) vs V2 Section 4 (Signal Scoring) vs V2 Section 6 (Knowledge Acquisition)

Conflict Description: V1.3 Section 3.1.2 assigns priorities: P1 (New Digital Bank License, CTO/CDO Hiring), P2 (Funding Round, Open Banking Initiative), P3 (Fintech Saudi Initiative). V2 Section 4 assigns points: P1 = 25, P2 = 15, P3 = 8. But V2 Section 6 (Knowledge Acquisition Engine) treats each bank as a 'living causal model' where signal significance depends on the bank's current state — a Funding Round (P2) for a bank about to issue an RFP is more valuable than a CTO Hiring (P1) at a bank with no budget cycle.

Rule A: V1.3 / V2 Section 4: Static priority-to-points mapping.

Rule B: V2 Section 6: Context-dependent signal significance.

Engine Impact: Static points ignore strategic context. The engine would prioritize a P1 signal at a locked-down account over a P2 signal at an account with an active buying window, violating Section 1's timing-over-volume principle.

Recommended Resolution: RESOLVE BY: Points are BASE values. Context modifiers from Section 6's causal model multiply the base: Aligned_to_active_initiative x 1.5, Counter_to_known_strategy x 0.3, Neutral x 1.0. The modifier is computed by the Knowledge Acquisition Engine and stored on the signal record as context_multiplier.

[MEDIUM] COH-B06: Weight Floor: SCORE-RECAL-002 Minimum vs Learning Loop Freedom

Sections: V2 Section 4 (SCORE-RECAL-002) vs V2 Section 13 (Organizational Learning Governance)

Conflict Description: SCORE-RECAL-002 says 'Signal Strength weight must NEVER drop below 25%. It is always the primary driver.' V2 Section 13 (Learning Governance) establishes that learning loops should optimize based on actual conversion data, and Section 13's Constitutional Constraints say Section 1 Meta-Principles are non-negotiable — but the 25% floor is an operational rule in Section 4, not a meta-principle in Section 1.

Rule A: SCORE-RECAL-002: Signal Strength $\geq$ 25% always.

Rule B: V2 Section 13: Learning loops optimize weights based on conversion data.

Engine Impact: If conversion data shows Signal Strength correlates poorly with meetings (e.g., Relationship Warmth is 3x more predictive), the 25% floor forces the engine to over-weight a weak predictor. This directly conflicts with Section 13's anti-Goodhart governance.

Recommended Resolution: RESOLVE BY: The floor is a SAFETY BOUND, not a performance constraint. Reframe: 'Signal Strength weight may not drop below 25% without A4 governance approval. If learning data shows Signal Strength correlation $<$ 0.3, flag for human review rather than auto-adjusting.' This preserves the guard rail while allowing governed adaptation.

CATEGORY C: STATE MACHINE AND LIFECYCLE CONFLICTS

State management is the coordination layer that prevents duplicate outreach, workflow collisions, and CRM desynchronization. Conflicts here cause the most visible operational failures.

[CRITICAL] COH-C01: Account State 'HOT' vs Account Tier 'HOT': Name Collision, Different Semantics

Sections: V1.3 Section 23.2 (Account States) vs V2 Section 5 (Tier Assignment) vs V1.3 Section 4.4 (Tier Logic)

Conflict Description: V1.3 Section 23.2 lists account states: WATCHING, ENRICHING, HOT, OUTREACH_ACTIVE, ENGAGED, MEETING_BOOKED, OPPORTUNITY, CUSTOMER, DORMANT. Separately, V2 Section 5 assigns tiers: HOT/WARM/COLD based on the composite score. The state 'HOT' and the tier 'HOT' are DIFFERENT CONCEPTS sharing the same name. State HOT means 'enriched and ready for outreach.' Tier HOT means 'score >= 75 with fresh signals.' An account can be Tier HOT but State WATCHING (not yet enriched). It can be State HOT but Tier WARM (score dropped below 75 after enrichment).

Rule A: V1.3 Section 23.2: State 'HOT' = enriched, ready for outreach (a lifecycle stage).

Rule B: V2 Section 5: Tier 'HOT' = score >= 75 with fresh signals (a scoring classification).

Engine Impact: Engineers will conflate these. A query like 'get all HOT accounts' is ambiguous. Workflow triggers that fire on 'account becomes HOT' will fire on the wrong event. State-based routing and tier-based routing interfere with each other.

Recommended Resolution: RESOLVE BY: Rename the account state from 'HOT' to 'OUTREACH_READY' to eliminate the name collision. The lifecycle becomes: WATCHING -> ENRICHING -> OUTREACH_READY -> OUTREACH_ACTIVE -> ENGAGED -> MEETING_BOOKED -> OPPORTUNITY -> CUSTOMER -> DORMANT. Tier (HOT/WARM/COLD) and State are independent dimensions tracked on the Account entity.

[CRITICAL] COH-C02: Missing State Transitions: What Happens When a HOT Account's Score Drops to WARM?

Sections: V2 Section 5 (Tier Downgrade) vs V1.3 Section 23 (State Machine) vs V1.3 Section 7 (Outreach Engine)

Conflict Description: V2 Section 5 specifies tier downgrade rules (HOT -> WARM when score <= 70). V1.3 Section 7 says an active outreach sequence runs Touches at Day 1/4/8/14. But what happens if an Account downgrades from HOT to WARM while an outreach sequence is mid-execution (e.g., after Touch 2 on Day 4)? Neither section specifies this. The state machine has no transition for OUTREACH_ACTIVE + tier_downgrade.

Rule A: V2 Section 5: Tier can downgrade from HOT to WARM at any time.

Rule B: V1.3 Section 7: Outreach sequence runs for 14 days with 4 touches. V1.3 Section 23: No state transition defined for mid-outreach tier change.

Engine Impact: The engine either (a) continues outreach on a now-WARM account (wasting capital on a deprioritized target), (b) abruptly stops the sequence (leaving the prospect mid-conversation), or (c) undefined behavior (depends on implementation). All three are wrong.

Recommended Resolution: RESOLVE BY: Add TIER-DOWN-SEQ-001: 'If an Account downgrades from HOT to WARM while OUTREACH_ACTIVE, the current sequence completes but no new sequence is initiated. If the Account downgrades while in ENRICHING, enrichment completes but outreach is not triggered. If the downgrade reaches COLD, all non-human interactions are paused immediately.' Also add V2 Section 5 override: 'Active engagement (REPLIED status on any Contact) blocks tier downgrade — ENGAGED accounts cannot be downgraded.'

[HIGH] COH-C03: Contact State REPLIED vs Account State ENGAGED: Trigger Ambiguity

Sections: V1.3 Section 23.3 (Contact States) vs V1.3 Section 23.4 (State Transition) vs V1.3 Section 8 (Engagement)

Conflict Description: V1.3 Section 23.4 shows: 'Reply Received -> Contact State REPLIED -> Pause Automation -> Account State ENGAGED -> Trigger Sales Handoff.' But V1.3 Section 8.2 lists 'Reply Received' as an engagement event that triggers 'Decision: Positive Intent? YES -> Trigger Human Handoff. NO -> Continue Monitoring.' These two specifications CONFLICT: Section 23.4 says ALL replies trigger handoff. Section 8 says only POSITIVE replies do.

Rule A: V1.3 Section 23.4: Reply -> ENGAGED -> Sales Handoff (unconditional).

Rule B: V1.3 Section 8: Reply -> Positive Intent check -> Handoff only if YES.

Engine Impact: If Section 23.4 is implemented, negative replies ('not interested') trigger unnecessary sales handoffs. If Section 8 is implemented, the state machine skips ENGAGED for negative replies, leaving the account in OUTREACH_ACTIVE with a REPLIED contact — an inconsistent state.

Recommended Resolution: RESOLVE BY: Section 8's positive-intent check is correct. The state transition should be: Reply Received -> Contact State REPLIED -> Pause Automation -> AI Sentiment Analysis -> IF positive: Account State ENGAGED + Trigger Handoff. IF negative/neutral: Account State remains OUTREACH_ACTIVE, Contact moves to DISQUALIFIED or continues in sequence. Update Section 23.4 to match Section 8's intent-check logic.

[HIGH] COH-C04: DORMANT Re-activation: No Return Path Defined

Sections: V1.3 Section 23.2 (Account States) vs V1.3 Section 3 (Signal Detection) vs V2 Section 16 (Temporal Intelligence)

Conflict Description: The state machine includes DORMANT as a terminal-adjacent state but defines no exit transitions. V1.3 Section 3 says new signals trigger scoring regardless of account state. V2 Section 16 (Temporal Intelligence) specifies timing windows and decay patterns. If a DORMANT account receives a strong new signal (new CTO hire at a previously cold bank), should it re-enter WATCHING? Or is DORMANT sticky?

Rule A: V1.3 Section 23.2: DORMANT listed with no exit transitions.

Rule B: V1.3 Section 3: Signals trigger scoring regardless. V2 Section 16: Temporal decay patterns imply renewal.

Engine Impact: DORMANT accounts with new high-value signals are either (a) trapped in DORMANT forever (data loss), or (b) silently re-activated without governance (unpredictable behavior).

Recommended Resolution: RESOLVE BY: Add STATE-DORMANT-001: 'DORMANT accounts re-enter WATCHING when a new P1 or P2 signal is ingested AND the resulting composite score exceeds the WARM threshold (50). The re-activation event is logged with reason and the previous dormancy period. Re-activated accounts receive a DORMANT_HISTORY tag that informs outreach tone (do not reference the gap in contact explicitly).'

[MEDIUM] COH-C05: Account State OPPORTUNITY vs Opportunity Entity: Circular Definition

Sections: V1.3 Section 23.2 (Account State 'OPPORTUNITY') vs V2 Section 1 (Entity 38 'Opportunity')

Conflict Description: V1.3 Section 23.2 has an account state called 'OPPORTUNITY' which means 'CRM opportunity created.' V2 Section 1 has an entity called 'Opportunity' (E38) which is the unit of selling. The account state 'OPPORTUNITY' and the entity 'Opportunity' are related but not identical — the state is a lifecycle marker, the entity is a data object. An Account can be in state OPPORTUNITY without an E38 entity existing (if CRM opportunity was created manually), and an E38 entity can exist before the Account reaches state OPPORTUNITY (if the engine creates the internal Opportunity early, per COH-A04 resolution).

Rule A: V1.3 Section 23.2: Account state 'OPPORTUNITY' (lifecycle marker).

Rule B: V2 Section 1: Entity 38 'Opportunity' (data object, unit of selling).

Engine Impact: State and entity are coupled by name but not by lifecycle. Engineers will assume they're synchronized, creating bugs where the state changes but the entity doesn't exist or vice versa.

Recommended Resolution: RESOLVE BY: Rename account state 'OPPORTUNITY' to 'DEAL_ACTIVE' to disambiguate from the Opportunity entity. Add invariant: 'Account state DEAL_ACTIVE requires at least one Opportunity entity in status OPEN or NEGOTIATING. An Account cannot enter DEAL_ACTIVE without a corresponding E38 record.'

CATEGORY D: AI AGENT AND AUTONOMY CONFLICTS

The AI layer sits between intelligence and execution. Conflicts between the multi-agent architecture, the Autonomy Ladder, and the human-in-the-loop requirements create the most dangerous category of silent failures.

[CRITICAL] COH-D01: MVP 'Review All AI Output' vs Autonomy Ladder A2 (Autonomous with Audit)

Sections: V1.3 Section 16.4 (Operational Principles) vs V2 Section 5 GOV (Autonomy Ladder) vs V1.3 Section 15.7

Conflict Description: V1.3 Section 16.4 says 'Review all AI output initially.' V1.3 Section 15.7 says 'Sending without human review = Enterprise brand risk. Manual approval during MVP.' But V2's Autonomy Ladder defines A2 as 'Autonomous with Audit' — the system acts and logs for review, not requiring pre-approval. Several module operations in V2 are classified as A2. If MVP requires manual pre-approval of ALL AI output, then nothing can operate at A2 or above during MVP.

Rule A: V1.3: ALL AI output requires human review during MVP (pre-approval).

Rule B: V2 GOV: A2 actions are autonomous with post-hoc audit (no pre-approval).

Engine Impact: This either (a) makes the Autonomy Ladder meaningless during MVP (everything is A3+ regardless of classification), or (b) creates confusion about which rule takes precedence. The 'initially' in V1.3 is undefined — when does the 'initially' period end?

Recommended Resolution: RESOLVE BY: Define MVP_REVIEW_PHASE as weeks 1-4 (first month). During MVP_REVIEW_PHASE, all operations classified A2 or above are elevated to A3 (human pre-approval required). After MVP_REVIEW_PHASE, operations return to their classified autonomy level IF error rate < 5% during the review phase. Add GOV-MVP-001: 'The MVP review period overrides autonomy classifications. Phase exit requires: (a) 4 weeks elapsed, (b) < 5% human override rate on A2 actions, (c) explicit human authorization to exit the phase.'

[HIGH] COH-D02: QC Agent Temperature 0.1 vs Message Generation Temperature 0.4: QC Can Reject Valid Messages

Sections: V2 Section 7 (AI Agent Architecture) vs V2 Section 11 (Human-AI Decision Partnership)

Conflict Description: V2's AI agent architecture specifies Message Generation Agent at temperature 0.4 and QC Agent at temperature 0.1. The QC Agent validates 'factual grounding, tone, enterprise suitability, personalization quality, compliance risk.' But a 0.1-temperature QC Agent with strict validation criteria may reject messages that a 0.4-temperature generator produces as appropriately creative. The QC Agent's rejection triggers regeneration — but regeneration at the same temperature will produce similar output, creating a potential infinite retry loop.

Rule A: V2 Section 7: Message Gen at 0.4 (moderate creativity). QC at 0.1 (very strict).

Rule B: V2 Section 11: Human-AI Partnership requires the system to produce output humans can act on.

Engine Impact: Retry loops between generation and QC. Each retry burns API cost and time. If QC criteria are too strict for the generation temperature, the system produces nothing or produces the blandest possible message (which fails Section 1's personalization-over-volume principle).

Recommended Resolution: RESOLVE BY: Add QC-RETRY-001: 'Maximum 3 regeneration attempts per message. After 3 QC rejections, escalate to A3 (human review) with all three rejected drafts and QC rejection reasons attached. On the 2nd retry, generation temperature drops to 0.3. On the 3rd, drops to 0.2.' Also: QC rejection reasons must be structured (not free-text) to enable prompt optimization in Section 13's learning loop.

[HIGH] COH-D03: n8n Zero Business Logic (ORCH-NOLOGIC-001) vs Workflow Decision Nodes

Sections: V2 Section 3 (ORCH-NOLOGIC-001) vs V1.3 Section 21 (n8n Orchestration) vs V1.3 Section 3.1.4 (Signal Decision Logic)

Conflict Description: ORCH-NOLOGIC-001 says n8n contains zero business logic — 'business logic, scoring logic, AI reasoning logic belong in APIs/services.' But V1.3 Section 3.1.4 shows decision nodes INSIDE workflows: 'Decision Node: Is Signal Duplicate? IF signal already exists within freshness window -> Ignore signal.' V1.3 Section 7.3 shows 'Decision: Reply Received? YES -> Pause Automation.' These are business logic decisions implemented as n8n workflow nodes, directly violating ORCH-NOLOGIC-001.

Rule A: V2: ORCH-NOLOGIC-001 — n8n contains zero business logic.

Rule B: V1.3: Decision nodes in n8n workflows make business logic decisions (duplicate check, reply routing).

Engine Impact: If ORCH-NOLOGIC-001 is enforced, all decision nodes must call APIs (adding latency and failure points to every workflow step). If V1.3's approach is kept, ORCH-NOLOGIC-001 is violated and business logic is scattered across n8n JSON files and API code — making the system unauditable.

Recommended Resolution: RESOLVE BY: Refine ORCH-NOLOGIC-001 to ORCH-NOLOGIC-002: 'n8n may contain ROUTING logic (if/else based on API response codes or data presence checks) but not COMPUTATION logic (scoring, threshold evaluation, sentiment analysis, weight calculations). The test: if the decision requires reading a business threshold from the Rule_Registry, it belongs in the API. If it checks whether an API returned data or not, it can live in n8n.' Duplicate detection MOVES to the Signal API. Reply routing stays in n8n (it checks a webhook payload field, not a business threshold).

[MEDIUM] COH-D04: Hallucination Prevention: Never Invent vs Structured Reasoning Requirements

Sections: V1.3 Section 22.5 (Hallucination Prevention) vs V2 Section 6 (Knowledge Acquisition) vs V2 Section 10 (Reality Calibration)

Conflict Description: V1.3 Section 22.5 lists absolute prohibitions: 'AI Must NEVER invent funding rounds, partnerships, product launches, regulatory approvals, initiatives.' V2 Section 6 (Knowledge Acquisition Engine) and V2 Section 10 (Reality Calibration) require the AI to build hypotheses about account priorities, infer pain points, and predict buying triggers — all of which are forms of 'inventing' inferences about unverified realities.

Rule A: V1.3: NEVER invent facts about accounts.

Rule B: V2 Section 6/10: Build hypotheses, infer pain, predict triggers (creative inference required).

Engine Impact: The prohibition is too broad. 'Never invent a funding round' is correct — that's a factual claim. But 'this CTO's hire suggests a modernization initiative' is a HYPOTHESIS, not an invention. The current language makes all inference look like hallucination.

Recommended Resolution: RESOLVE BY: Split the prohibition. HALLUC-FACTUAL-001: 'AI must never assert as fact any event, number, date, partnership, or initiative not grounded in a source with EPIS confidence ≥ 0.6 .' HALLUC-INFER-001: 'AI may generate HYPOTHESES about account priorities, pain points, and buying triggers, but must label them as hypotheses (H:) and attach the evidence chain that supports the inference. Hypotheses are never presented to prospects as facts — they inform outreach strategy, not outreach content.'

CATEGORY E: TIMING AND SEQUENCE CONFLICTS

Timing is the system's competitive weapon (Section 1, Principle 4). Conflicts in timing rules directly erode the core value proposition.

[HIGH] COH-E01: Email Sequence Day 1/4/8/14 vs LinkedIn Day 1/+2/+7: Channel Collision

Sections: V1.3 Section 7.4 (Email Sequence) vs V1.3 Section 7.4 (LinkedIn Sequence) vs V1.3 Section 15.7 (Pitfall: Simultaneous Channel Spam)

Conflict Description: V1.3 Section 7.4 Email: Touch 1 Day 1, Touch 2 Day 4, Touch 3 Day 8, Touch 4 Day 14. LinkedIn: Step 1 Day 1, Step 2 +2 Days, Step 3 +7 Days. V1.3 Section 7.3 shows LinkedIn launches AFTER the email engagement window. But the timing tables show both starting Day 1. V1.3 Section 15.7 explicitly warns: 'Simultaneous channel spam = Poor prospect experience. Coordinate email + LinkedIn timing.'

Rule A: V1.3 Section 7.4: Both email and LinkedIn start Day 1.

Rule B: V1.3 Section 7.3 Flow: LinkedIn launches after email engagement window. V1.3 Section 15.7: Don't spam simultaneously.

Engine Impact: Three specifications, three different answers for when LinkedIn starts. Day 1 (timing table), after email engagement window (flow diagram), or 'coordinated' (pitfall warning — undefined).

Recommended Resolution: RESOLVE BY: LinkedIn connection request sends on Day 1 (low-cost, non-intrusive). LinkedIn value DM sends on Day 5 (after Email Touch 2 on Day 4 — different day, reinforcing signal). LinkedIn contextual follow-up on Day 10 (between Email Touch 3 Day 8 and Touch 4 Day 14). The channels INTERLEAVE, never overlap on the same day. Add EXEC-CHANNEL-001: 'No two outreach touches on the same Contact on the same calendar day.'

[HIGH] COH-E02: Send Window Tue-Thu 9-11am vs KSA Work Week Sun-Thu

Sections: V2 Section 9 (Outreach Cadence) vs V2 Section 17 (KSA-Specific Rules) vs V1.3 Section 7.4

Conflict Description: V2 Section 9 specifies optimal send windows as Tuesday-Thursday 9-11am. But V2 Section 17 (KSA rules) establishes the KSA work week as Sunday-Thursday, with Friday-Saturday as the weekend. Sunday and Monday are valid KSA work days excluded from the Tue-Thu window. Tuesday-Thursday 9-11am is only 3 of 5 KSA work days — 40% of the available sending window is blocked by a rule calibrated for Western work weeks.

Rule A: V2 Section 9: Send emails Tue-Thu 9-11am.

Rule B: V2 Section 17: KSA work week is Sun-Thu. Friday-Saturday is weekend.

Engine Impact: The engine under-utilizes 40% of the KSA work week. Sunday and Monday outreach is never attempted, missing two valid business days. The Tue-Thu window is a Western-market default, not KSA-optimized.

Recommended Resolution: RESOLVE BY: Replace with KSA-optimized send window: SEND-KSA-001: 'Primary send window: Sun-Wed 9:00-11:00 AM AST. Thursday is a light work day (reduced hours) — send before 12:00 PM only. Friday-Saturday: no sends.' This gives 4.5 sending days instead of 3, matching the actual KSA business rhythm.

[MEDIUM] COH-E03: Signal Polling Frequency: 'Every X Hours' Undefined

Sections: V1.3 Section 3.1.3 (Signal Detection Workflow) vs V2 Section 16 (Temporal Intelligence)

Conflict Description: V1.3 Section 3.1.3 says 'Cron Trigger (Every X Hours)' — X is undefined. V1.3 Section 15.2 says 'Run scheduled polling frequently' (also undefined). V2 Section 16 (Temporal Intelligence) specifies timing windows for engagement but not for signal polling. Meanwhile, V1.3 Section 15.2 warns 'Delayed signal processing = Missed timing opportunities.'

Rule A: V1.3 Section 3.1.3: Signal polling frequency = 'Every X Hours' (X undefined).

Rule B: V2 Section 16: No polling frequency specified. V1.3 Section 15.2: Poll 'frequently' (undefined).

Engine Impact: An engineer will pick an arbitrary frequency. Too frequent (every 15 min) wastes API calls. Too infrequent (every 24 hours) violates timing-over-volume and misses breaking signals (CTO hire announced at 2pm, competitor reaches out by 5pm).

Recommended Resolution: RESOLVE BY: Specify per-source polling: SIG-POLL-001: 'Google News API: every 4 hours. SAMA RSS: every 6 hours. LinkedIn (via enrichment): every 12 hours. Crunchbase: every 24 hours. Job boards: every 12 hours. Company websites: weekly.' The frequency matches the source's typical update cadence and the signal's time-sensitivity (regulatory signals are less frequent but critical; news is frequent and time-sensitive).

[MEDIUM] COH-E04: Ramadan/Eid Blackout: Total Silence vs Continued Monitoring

Sections: V2 Section 17 (KSA Rules - Ramadan) vs V2 Section 3 (Signal Monitoring) vs V2 Section 8 (Trust Preservation)

Conflict Description: V2 Section 17 specifies Ramadan and Eid as blackout periods for outreach. But V2 Section 3 (Signal Monitoring) says the system 'continuously observes external market activity.' V2 Section 8 (Trust Preservation) says relationship momentum must be maintained. If outreach is blacked out but signals continue, HOT accounts accumulate unactioned signals. When the blackout ends, the engine faces a backlog of stale signals that were once fresh.

Rule A: V2 Section 17: No outreach during Ramadan/Eid.

Rule B: V2 Section 3/8: Continuous monitoring + relationship momentum maintenance.

Engine Impact: Post-Ramadan signal backlog: signals detected during Ramadan are 30+ days old by Eid. The fresh-signal gate (V2 Section 5) would cap these accounts at WARM, even though the underlying opportunity is real.

Recommended Resolution: RESOLVE BY: SIG-RAMADAN-001: 'Signal ingestion and scoring continue during Ramadan/Eid. Outreach execution is paused. Signals detected during Ramadan have their freshness clock paused (age frozen at detection time). On Eid+1, freshness clocks resume from frozen age. Accounts that crossed HOT during Ramadan are queued for Day 1 post-blackout outreach in priority order.' This preserves timing intelligence while respecting the cultural blackout.

CATEGORY F: INTEGRATION AND DATA FLOW CONFLICTS

[CRITICAL] COH-F01: Source of Truth: PostgreSQL vs HubSpot CRM — Who Wins on Conflict?

Sections: V1.3 Section 15.9 (CRM Pitfalls) vs V1.3 Section 19 (DB Architecture) vs V2 Section 12 (Distributed Reality Coordination)

Conflict Description: V1.3 Section 15.9 says 'Use HubSpot as central CRM' — implying HubSpot is the source of truth. V1.3 Section 19 says the database is 'Source of truth, Intelligence memory layer, Workflow coordination layer.' V2 Section 12 (Distributed Reality Coordination) introduces INT-CRM as one of 12 integration domains with its own EPIS confidence scores. If a sales rep updates an Account's relationship_status to 'Strong' in HubSpot, but the engine's scoring shows relationship_warmth = 15 (weak), which record wins?

Rule A: V1.3 Section 15.9: HubSpot = central CRM (implied source of truth).

Rule B: V1.3 Section 19: PostgreSQL = source of truth. V2 Section 12: CRM is one integration domain, not the authority.

Engine Impact: Every CRM sync is a potential data conflict. If PostgreSQL wins, sales team updates are overwritten (they stop trusting the system). If HubSpot wins, the engine's intelligence is overridden by manual entries (losing the AI advantage).

Recommended Resolution: RESOLVE BY: Split authority by field type. INT-CRM-AUTHORITY-001: 'For RELATIONSHIP fields (owner, status, notes, next_action): HubSpot is authoritative — human relationship knowledge takes precedence. For INTELLIGENCE fields (score, tier, signal history, AI research): PostgreSQL is authoritative — the engine's analysis takes precedence. For SHARED fields (contact info, company data): last-write-wins with conflict flag and 24-hour reconciliation review.' This mirrors the AI-assists-humans principle: humans own relationships, AI owns intelligence.

[HIGH] COH-F02: Enrichment Sources: Apollo + Clearbit + Clay vs EPIS Confidence Requirements

Sections: V1.3 Section 5.2 (Enrichment Sources) vs V2 Section 4 (EPIS Reality Confidence Model) vs V1.3 Section 15.4

Conflict Description: V1.3 Section 5.2 lists enrichment sources: Apollo (contacts + emails), Clearbit (firmographic), Clay (workflow orchestration), LinkedIn (activity + roles). V2 Section 4 establishes EPIS (Evidence, Provenance, Independence, Staleness) as the confidence model for all data. But V1.3 assigns no EPIS scores to enrichment sources. Is Apollo a high-confidence source? Is Clay-orchestrated data trustworthy? V1.3 Section 15.4 warns 'Outdated enrichment data = Low personalization relevance.'

Rule A: V1.3: Enrichment sources listed without confidence assessment.

Rule B: V2: EPIS confidence model governs all data. No EPIS scores assigned to enrichment sources.

Engine Impact: The engine treats all enrichment data equally. An Apollo-sourced email with 90% deliverability score and a Clay-scraped email with unknown provenance both enter the system as 'verified.' The QC Agent cannot distinguish reliable from unreliable enrichment.

Recommended Resolution: RESOLVE BY: Assign base EPIS profiles to each source: ENR-EPIS-001: 'Apollo contacts: E=0.8, P=0.9 (established provider), I=0.7, S=per-record. Clearbit firmographic: E=0.7, P=0.8, I=0.6, S=per-record. Clay-orchestrated: E=0.5, P=0.5 (depends on sub-sources), I=0.4, S=per-record. LinkedIn: E=0.9, P=0.95, I=0.8, S=per-record.' These profiles feed the EPIS-RCM stamp on every enrichment record.

[HIGH] COH-F03: Signals 'Should NEVER Be Overwritten' vs Score Recalculation on Enrichment Update

Sections: V1.3 Section 19.4 (Signals Design Principle) vs V2 Section 4 (SCORE-CALC-003)

Conflict Description: V1.3 Section 19.4 says 'Signals should NEVER be overwritten. All signals should remain historically queryable.' SCORE-CALC-003 says score recalculates on 'enrichment data updated.' But when enrichment updates firmographic data that changes the Regulatory Pressure dimension (e.g., discovering the bank has a new SAMA compliance deadline), should this create a NEW signal or UPDATE existing data? If it creates a new signal, the signal table grows unboundedly with enrichment-derived entries that aren't real 'market signals.' If it updates existing data, it violates the immutability principle.

Rule A: V1.3 Section 19.4: Signals are append-only, never overwritten.

Rule B: V2 Section 4: Score recalculates on enrichment updates (implies enrichment can affect signal-like data).

Engine Impact: Enrichment data and signals are conflated. The scoring engine treats them differently (signals have freshness decay, enrichment doesn't), but the trigger to rescore doesn't distinguish the source.

Recommended Resolution: RESOLVE BY: Separate signal and enrichment data paths. SCORE-CALC-005: 'Score recalculation triggers are classified: (a) SIGNAL triggers — new market signals, subject to freshness decay, drive Signal Strength dimension; (b) ENRICHMENT triggers — firmographic/contact updates, no freshness decay, drive Reachability and Relationship dimensions; (c) ENGAGEMENT triggers — prospect interactions, time-weighted, drive Relationship Warmth. Each trigger type affects only its designated dimensions.' Enrichment updates are stored in the enrichment table (append-only), not the signals table.

[MEDIUM] COH-F04: Email Tool: Instantly vs Smartlead — No Selection Criteria

Sections: V1.3 Section 7.2 (Supported Channels) vs V1.3 Section 11 (Technology Stack)

Conflict Description: V1.3 Section 7.2 lists 'Email: Instantly / Smartlead' with a slash implying either/or. V1.3 Section 11 lists 'Email Execution: Instantly / Smartlead' the same way. No selection criteria are specified. Both are listed in parallel with no guidance on when to use which, whether both are used simultaneously, or how to handle domain reputation across two providers.

Rule A: V1.3: Two email tools listed with no selection criteria.

Rule B: V2: No guidance on email provider selection or multi-provider strategy.

Engine Impact: An engineer will pick one arbitrarily. If both are used, domain reputation is split across providers (harder to manage). If one is picked, the other listing creates confusion.

Recommended Resolution: RESOLVE BY: Pick one for MVP and specify why. EXEC-EMAIL-001: 'MVP uses Instantly as the sole email execution provider. Smartlead is the designated failover. Reasons: [specify based on actual evaluation]. Multi-provider operation is deferred to post-MVP with a domain reputation unification strategy.' Remove the slash-notation from all documentation.

CATEGORY G: GOVERNANCE AND LEARNING LOOP CONFLICTS

[CRITICAL] COH-G01: Constitutional Constraint (Section 13) vs Analytics Weight Update (Section 10)

Sections: V2 Section 13 (Constitutional Constraints) vs V1.3 Section 10.3 (Analytics Workflow) vs V2 Section 4 (SCORE-RECAL)

Conflict Description: V2 Section 13 establishes that Section 1 Meta-Principles are non-negotiable — 'no learning loop can override them.' V1.3 Section 10.3 Analytics Workflow shows: 'Update Scoring Weights -> Improve AI Prompts' as automated steps. V2 Section 4 SCORE-RECAL-001 specifies weight adjustment criteria. But the analytics workflow in V1.3 Section 10.3 shows weight updates as automated, with no governance gate. If the learning loop auto-adjusts weights that violate a meta-principle (e.g., reducing Signal Strength below the principle that timing/signals are primary), the Constitutional Constraint is violated without detection.

Rule A: V2 Section 13: Meta-Principles cannot be overridden by learning loops.

Rule B: V1.3 Section 10.3: Analytics auto-updates scoring weights. No governance gate visible.

Engine Impact: The learning loop can silently erode meta-principles through incremental weight changes that individually seem small but cumulatively undermine a core principle. This is the Goodhart's Law failure that Section 13 v2 was explicitly designed to prevent.

Recommended Resolution: RESOLVE BY: Every weight update must pass through a Constitutional Constraint Checker before application. GOV-CONST-001: 'No automated learning loop may modify any parameter that, when changed, would cause a meta-principle evaluation to return FALSE. The checker runs the 25 meta-principles against the proposed new parameters before applying them. Violations are blocked and routed to A4 governance review.' The analytics workflow in Section 10.3 must include this gate between 'Analyze' and 'Update.'

[HIGH] COH-G02: MVP KPI Targets vs Statistical Significance Requirements

Sections: V2 Section 13 (KPI Targets) vs V2 Section 4 (SCORE-RECAL-001) vs V2 Section 13 (Anti-Goodhart Governance)

Conflict Description: V2 Section 13 specifies MVP KPI targets: 4-8 meetings/month, >8% reply rate, >3% positive reply, >40% open rate. V2 Section 4 SCORE-RECAL-001 requires '10 meetings booked' before recalibration. At 4-8 meetings/month, reaching 10 meetings takes 1.5-2.5 months. But V2 Section 13's anti-Goodhart governance warns against optimizing on small samples. With only 10-20 data points, any 'correlation analysis' for weight recalibration is statistically unreliable.

Rule A: V2 Section 13: 4-8 meetings/month as MVP target. V2 Section 4: 10 meetings as recalibration threshold.

Rule B: V2 Section 13: Anti-Goodhart warning against small-sample optimization.

Engine Impact: The system recalibrates weights based on 10 data points — below any reasonable significance threshold. The recalibrated weights may reflect noise rather than signal, and the anti-Goodhart governance designed to prevent exactly this has no concrete enforcement mechanism.

Recommended Resolution: RESOLVE BY: Raise the recalibration threshold. SCORE-RECAL-001-REVISED: 'Weight recalibration requires minimum 30 meetings booked AND 90 days of operation AND p-value < 0.05 on dimension-conversion correlation. Below these thresholds, weights remain at initial values. Pre-threshold period uses A/B testing with control group (10% of outreach uses randomized weights) to build the statistical foundation for recalibration.' This enforces Section 13's anti-Goodhart principle concretely.

[MEDIUM] COH-G03: Prompt Versioning (Section 15.6) vs AI Model Registry (V2 Entity 27) — Redundant Governance

Sections: V1.3 Section 15.6 (Prompt Engineering Pitfalls) vs V2 Section 2 (Entity 27: AI_Model_Registry) vs V1.3 Section 19.6

Conflict Description: V1.3 Section 15.6 says 'Store prompts in database. Use prompt versioning.' V1.3 Section 19.6 describes the AI Generations Table storing 'prompts, outputs, model used, QC results, hallucination scores, approval status.' V2 Section 2 defines Entity 27 (AI_Model_Registry) which catalogs AI models with versions, capabilities, and performance. But prompt versions and model versions are tracked in different entities with no explicit link. A prompt that works with Claude 3.5 Sonnet may fail with GPT-4o — the prompt-model compatibility matrix is unspecified.

Rule A: V1.3 Section 19.6: Prompts stored with model_used per generation.

Rule B: V2 Entity 27: Models tracked separately with their own versioning.

Engine Impact: When a model is upgraded (e.g., Claude 3.5 -> Claude 4), all prompts must be re-tested. But there's no mechanism to flag which prompts need re-testing. Prompt version 5 + Model version A may produce different results than Prompt version 5 + Model version B.

Recommended Resolution: RESOLVE BY: Add PROMPT-MODEL-001: 'Each prompt version is tagged with a list of validated model versions. When a model is updated in AI_Model_Registry, all prompts validated only against the old model version are flagged for re-testing. Re-testing uses the QC Agent with a curated test set of 10 representative inputs. The prompt is only cleared for the new model version after passing QC on all 10.' Link the AI_Generations table to both prompt_version_id AND model_version_id.

CATEGORY H: KSA-SPECIFIC AND COMPLIANCE CONFLICTS

[HIGH] COH-H01: PDPL Compliance vs Enrichment Data Collection Scope

Sections: V1.3 Section 25.3 (Compliance) vs V1.3 Section 5 (Enrichment) vs V2 Section 4 (EPIS) vs V2 Section 17 (KSA Rules)

Conflict Description: V1.3 Section 25.3 says the platform should align with 'PDPL requirements.' V1.3 Section 5 specifies enriching contacts with emails, hiring trends, LinkedIn activity, firmographic data. V2 Section 17 will cover KSA-specific rules including PDPL. But the enrichment engine's data collection scope is defined WITHOUT reference to PDPL. Under Saudi PDPL, collecting and processing personal data of Saudi nationals requires legitimate interest or consent. Scraping LinkedIn profiles of Saudi banking executives and storing their data in a PostgreSQL database may violate PDPL Article 5 (consent requirement) and Article 10 (cross-border transfer restrictions — data stored on Railway/Render which may be hosted outside KSA).

Rule A: V1.3 Section 5: Enrich contacts from Apollo, LinkedIn, Clearbit (data collection).

Rule B: V1.3 Section 25.3: Align with PDPL. V2 Section 17: KSA rules (not yet fully specified).

Engine Impact: The enrichment engine may collect data in violation of PDPL. If a prospect files a PDPL complaint, Decimal's entire ABM operation in KSA is at regulatory risk — exactly the kind of risk Section 1's Legal/Compliance tier of the Constraint Hierarchy is designed to prevent.

Recommended Resolution: RESOLVE BY: This requires legal review. As a SAFE DEFAULT (per OPEN-Q governance): COMPL-PDPL-001: 'All contact enrichment for Saudi nationals must be flagged with `legitimate_interest_basis`. Acceptable bases: published LinkedIn profile (public data), conference speaker (public appearance), company website listing (corporate disclosure). Non-acceptable without consent: private email, personal phone, family connections. Data storage must be reviewed for KSA data residency requirements before production launch.'

[HIGH] COH-H02: Arabic Outreach Generation vs English-Only AI Architecture

Sections: V1.3 Section 6.5 (AI-003: Arabic outreach) vs V1.3 Section 22 (AI Architecture) vs V2 Section 7 (AI Agents)

Conflict Description: V1.3 AI-003 requires 'System shall generate Arabic outreach.' But V1.3 Section 22 and V2 Section 7 specify the AI agent architecture entirely in English — prompts, reasoning chains, QC validation criteria. Arabic generation requires: (1) Arabic-capable LLM, (2) Arabic QC validation (tone, formality, dialect — MSA vs Saudi dialect), (3) Arabic email subject line optimization, (4) Arabic LinkedIn messaging norms. None of these are specified in the AI architecture sections.

Rule A: V1.3 AI-003: Arabic outreach required.

Rule B: V1.3 Section 22 / V2 Section 7: English-only AI architecture.

Engine Impact: Arabic outreach will be either (a) not implemented (violating the requirement), (b) implemented as translation-only (poor quality — enterprise Arabic has specific formality registers), or (c) implemented ad-hoc without the architectural rigor applied to English outreach.

Recommended Resolution: RESOLVE BY: Defer Arabic to Phase 2 explicitly. AI-003-REVISED: 'Phase 1 (MVP): English outreach only. Phase 2: Arabic outreach with dedicated Arabic QC Agent, MSA formality validation, and Arabic-specific prompt chains. Transition gate: (a) Arabic-capable model benchmarked at enterprise-grade formality, (b) Arabic QC criteria validated by native speaker review.' Add Phase 2 flag to all sections that touch Arabic requirements.

[MEDIUM] COH-H03: Vision 2030 Alignment Requirement vs Outreach Content Rules

Sections: V2 Section 17 (KSA Rules) vs V1.3 Section 22.5 (Hallucination Prevention) vs V2 Section 7 (Message Generation)

Conflict Description: V2 Section 17 will specify Vision 2030 alignment as a KSA outreach requirement (mentioning digitization, financial inclusion, fintech enablement in outreach). V1.3 Section 22.5 prohibits inventing initiatives. But if the AI references a bank's Vision 2030 alignment without confirmed evidence, is that hallucination or contextual framing? Every KSA bank publicly claims Vision 2030 alignment, but specific initiatives (e.g., 'your digital lending transformation') may not be publicly confirmed.

Rule A: V2 Section 17: Reference Vision 2030 in outreach.

Rule B: V1.3 Section 22.5: Don't invent initiatives the bank hasn't announced.

Engine Impact: The AI either (a) references Vision 2030 generically (bland, not personalized), (b) references specific bank initiatives that may not exist (hallucination risk), or (c) avoids Vision 2030 entirely (missing the KSA framing that every local vendor uses).

Recommended Resolution: RESOLVE BY: OUTREACH-V2030-001: 'Vision 2030 references in outreach are permitted at two levels: (1) SECTOR-LEVEL (always safe): Reference Vision 2030's financial sector goals (these are public government documents). (2) BANK-LEVEL (evidence-required): Reference specific bank initiatives only when grounded in a signal with EPIS confidence ≥ 0.7 (annual report, press release, executive speech). Label in the prompt chain which level is being used.'

MASTER RESOLUTION PRIORITY

Not all conflicts are equal. The following priority order reflects engine-collapse risk, with the highest-risk conflicts first. This is the order in which resolutions must be confirmed before implementation can proceed.

Priority	Conflict ID	Severity	Category	Core Issue
1	COH-A01	CRITICAL	Anchor	Account vs Opportunity design — foundational
2	COH-B01	CRITICAL	Scoring	Two competing scoring models — gates everything
3	COH-A02	CRITICAL	Anchor	Score belongs to Account or Opportunity
4	COH-C01	CRITICAL	State	HOT state vs HOT tier name collision
5	COH-A03	CRITICAL	Anchor	Tier assignment for multi-Opportunity Accounts
6	COH-D01	CRITICAL	Autonomy	MVP review vs Autonomy Ladder — governs all AI
7	COH-C02	CRITICAL	State	Mid-outreach tier downgrade — no transition defined
8	COH-F01	CRITICAL	Integration	PostgreSQL vs HubSpot source of truth
9	COH-G01	CRITICAL	Governance	Constitutional constraint vs auto weight update
10	COH-B02	HIGH	Scoring	Hysteresis band vs weight recalibration
11	COH-B03	HIGH	Scoring	Three different freshness models
12	COH-C03	HIGH	State	Reply triggers handoff unconditionally vs intent check
13	COH-E01	HIGH	Timing	Email and LinkedIn channel collision
14	COH-E02	HIGH	Timing	Western send window vs KSA work week
15	COH-D02	HIGH	Autonomy	QC temp vs generation temp retry loop
16	COH-D03	HIGH	Autonomy	n8n zero logic vs workflow decision nodes
17	COH-A04	HIGH	Anchor	CRM Opportunity creation timing
18	COH-A05	HIGH	Anchor	Engagement events entity attribution
19	COH-F02	HIGH	Integration	Enrichment sources without EPIS scores
20	COH-F03	HIGH	Integration	Signal immutability vs enrichment rescoring
21	COH-B04	HIGH	Scoring	Scoring frequency and debounce
22	COH-C04	HIGH	State	DORMANT re-activation path missing
23	COH-G02	HIGH	Governance	Small-sample recalibration anti-Goodhart
24	COH-H01	HIGH	Compliance	PDPL vs enrichment data collection
25	COH-H02	HIGH	Compliance	Arabic outreach vs English-only AI architecture
26	COH-B05	MEDIUM	Scoring	Static signal priority vs contextual significance
27	COH-B06	MEDIUM	Scoring	Weight floor vs learning loop freedom
28	COH-C05	MEDIUM	State	Account state OPPORTUNITY vs Entity Opportunity
29	COH-D04	MEDIUM	Autonomy	Hallucination prohibition vs hypothesis generation
30	COH-E03	MEDIUM	Timing	Signal polling frequency undefined
31	COH-E04	MEDIUM	Timing	Ramadan blackout vs signal freshness
32	COH-F04	MEDIUM	Integration	Two email tools no selection criteria
33	COH-G03	MEDIUM	Governance	Prompt versioning vs model registry gap
34	COH-H03	MEDIUM	Compliance	Vision 2030 references vs hallucination rules

NEXT STEPS

WHAT THIS AUDIT ESTABLISHES

This audit identifies 34 cross-section conflicts across 8 categories. Nine of these are **CRITICAL** — meaning they will cause the engine to produce incorrect behavior, deadlock, or data corruption if left unresolved.

The three load-bearing conflict clusters are:

1. **The Anchor Model (COH-A01 through A05)** — The Account-vs-Opportunity tension runs through every section. Until this is resolved with concrete dual-entity specifications (Account for relationships, Opportunity for deals), every downstream section inherits the ambiguity.
2. **The Scoring Duality (COH-B01)** — Two scoring models (V1.3 composite vs V2 Effective Opportunity) with different scales, semantics, and entity anchors. Until the dual-score architecture is explicitly specified, Section 5 (tiers), Section 6 (enrichment triggers), and Section 7 (outreach decisions) cannot be coherently implemented.
3. **The State-Tier Confusion (COH-C01, C02)** — The name collision between state 'HOT' and tier 'HOT' combined with missing state transitions for mid-lifecycle tier changes will cause the most visible operational failures (duplicate outreach, dropped sequences, inconsistent CRM records).

Recommended approach: Resolve **CRITICAL** conflicts in the order listed in the Master Resolution Priority table. Each resolution produces new rule IDs that must be cross-referenced into all affected sections. **HIGH** conflicts should be resolved before production launch. **MEDIUM** conflicts can be resolved during engineering review.

AUDIT LIMITATIONS

This audit cross-references the V1.3 FSD base document and V2 deep-specification layers as captured in project knowledge and conversation history. The full rule-level content of the 20+ PDF artifacts produced in prior sessions is reconstructed from conversation summaries and code fragments — not from reading the actual PDF files (which must be uploaded per session for authoritative access). Specific rule IDs may have evolved between sessions. The audit identifies conflict **PATTERNS** that hold regardless of specific rule numbering.

V2 Sections 17-19 have not received deep-specification passes. Additional conflicts may surface as those sections are built. The KSA-specific conflicts (Category H) are preliminary — V2 Section 17's full specification will likely surface additional tensions.